

Lightweight Online Multivariate Conformal Prediction

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Abstract

In the context of Advanced Driver Assistance Systems (ADAS), autonomous planning modules require rigorous uncertainty quantification alongside trajectory forecasting to ensure safe and proactive decision-making. While Conformal Prediction (CP) provides statistically valid confidence guarantees, its extension to multivariate spaces remains under-explored and, when implemented, often results in significant computational overhead that limits real-time applicability. To address this, we propose a Lightweight Online Multivariate Conformal Prediction (L-OMCP) framework designed to operate as an efficient top layer for existing prediction modules. Our approach constructs 2D confidence regions by introducing a joint non-conformity measure based on the empirical aspect ratio of spatial prediction errors, maintaining the lightweight computational overhead necessary for online deployment. To handle non-stationary driving dynamics, the framework utilizes an Exponential Moving Average (EMA) as its online update mechanism integrated with Adaptive Conformal Inference (ACI). This coupled formulation is projected into a bounded angular phase space to ensure robust stabilization and mitigate the impact of transient outliers or numerical instabilities. Comprehensive evaluations on synthetic scenarios and the real-world Renault dataset demonstrate that the proposed framework maintains the target coverage. Compared to both independent 1D baselines and state of the art multivariate methods, our approach generates tighter, context-aware bounding regions, providing a solution for real-time uncertainty quantification in ADAS.

Keywords: Conformal Prediction, Multivariate, ADAS, Time Series Forecasting, Uncertainty Quantification

1. Introduction

In the rapidly evolving domain of Advanced Driver Assistance Systems (ADAS) and autonomous navigation, reliable trajectory forecasting is paramount for safe and proactive decision-making. While current machine learning and kinematic models excel at generating deterministic point predictions for dynamic objects (Zhou et al., 2023, 2022; Messaoud et al., 2025), these single-path outputs inherently lack rigorous uncertainty quantification. For an autonomous planning module to safely navigate complex environments, knowing a vehicle’s most likely future position is insufficient; the system requires strictly calibrated spatial confidence regions that probabilistically bound the true future trajectory.

To address the critical need for uncertainty quantification, Conformal Prediction (CP) has emerged as a highly promising framework (Vovk et al., 2005; Angelopoulos and Bates, 2022). The primary power of CP lies in its foundational properties: it is entirely model-agnostic, making it applicable to any underlying prediction algorithm (e.g., neural networks or kinematic-based model) without constraints and it is distribution-free, offering mathematically rigorous coverage guarantees without requiring prior assumptions about the underlying data distribution, unlike other uncertainty quantification methods. However, the standard CP framework relies heavily on the assumption of exchangeability among data points. In the context of ADAS and trajectory forecasting, data arrives as a sequential time series, fundamentally violating the exchangeability premise due to temporal dependencies. While significant modifications, often categorized under Conformal Forecasting (Stocker et al., 2025), have been introduced to handle non-exchangeable time-series data, the vast majority of these solutions are strictly formulated for univariate forecasting.

The transition from univariate Conformal Forecasting to applications requiring multidimensional outputs, such as 2D vehicle trajectories involving simultaneous longitudinal and lateral coordinates (x,y) , introduces a significant challenge. Applying 1D CP independently to each spatial dimension with Bonferroni correction (Stankeviciute et al., 2021) ignores the inherent spatial covariance, often yielding overly conservative or empirically invalid bounding boxes, particularly during complex maneuvers. While there is a clear necessity to expand univariate conformal forecasting into a multidimensional space—and eventually to multi-step horizons to build confidence sets for entire future trajectories—existing multivariate CP methods are typically computationally expensive (Xu et al., 2024). This computational burden severely restricts their deployment in real-time online ADAS modules, where lightweight latency is a strict operational constraint.

In this work, we address the critical expansion of Conformal Prediction to multivariate spaces (especially for the 2D case) while strictly adhering to the computational constraints of real-time ADAS applications. We propose a Lightweight Online Multivariate Conformal Prediction (L-OMCP) framework specifically designed to operate as an efficient top layer within existing trajectory prediction modules. By introducing a non-conformity score derived from joint feature of the spatial prediction errors, our approach intrinsically links the dimensional errors. This enables the construction of geometrically proportional and optimized 2D confidence regions with minimal computational overhead.

Furthermore, to ensure the framework remains tight under non-stationary driving dynamics and distribution shifts (e.g., sudden lane changes or velocity variations), we introduce a robust online adaptation mechanism. We employ an Exponential Moving Average (EMA) to dynamically update the empirical aspect ratio based on the most recent trajectory observations. Crucially, to stabilize this dynamic calibration process and prevent catastrophic geometric inflation caused by transient prediction outliers, we mathematically project the error ratio into a bounded angular phase space prior to aggregation.

In summary, the main contributions of this work can be summarized as follows:

- We formulate a lightweight 2D joint non-conformity score based on the aspect ratio of recent spatial errors, efficiently extending Conformal Prediction to multivariate data without high computational costs to ensure real-time online applicability.

- We propose a robust online adaptation mechanism utilizing an Angular Phase-Space EMA, effectively stabilizing the real-time calibration process against numerical instabilities and transient prediction outliers in non-stationary environments.
- We provide comprehensive empirical validation on both synthetic scenarios and the real-world Renault trajectory dataset, demonstrating that our proposed framework maintains target empirical coverage while generating tighter, context-aware prediction regions compared to both Bonferroni-corrected independent 1D baselines and state-of-the-art multivariate methods.

2. Background

This section establishes the foundation of the study by first reviewing related work and existing literature in the field. It then formally introduces the mathematical foundations of Conformal Prediction, details the adaptations required for sequential and non-stationary data, formalizes the concepts of probabilistic coverage, and strictly defines the evaluation metrics utilized to assess the validity and efficiency of the generated confidence regions.

2.1. Related Work

The development of our proposed framework intersects with three primary domains of research: sequential conformal prediction, multivariate uncertainty quantification, and trajectory forecasting in autonomous systems.

Conformal Forecasting and Multivariate Extensions: Standard Conformal Prediction (Angelopoulos and Bates, 2022; Vovk et al., 2005) provides distribution-free bounds under the strict assumption of data exchangeability. To address the inherent temporal dependencies and distribution shifts in time-series forecasting, significant advancements have been made under the umbrella of Conformal Forecasting (Conformal Prediction for Time Series (Stocker et al., 2025)). As classified by (Stocker et al., 2025), these sequential methodologies restore the target coverage through four core philosophies: *Reweighting* (e.g., WCP / Nex-CP) (Barber et al., 2023), which assigns higher importance to historical points deemed more relevant to the current test point; *Refreshing*, which actively updates the calibration set via sliding windows to discard stale residuals (e.g., Enbpi) (Xu and Xie, 2023a); *Adapting Coverage*, which dynamically updates the target error rate online using feedback loops (e.g., ACI) (Gibbs and Candes, 2021; Zaffran et al., 2022); and *Blocking*, which redefines the unit of randomization by permuting entire blocks of data as exchangeable units rather than individual points (Chernozhukov et al., 2018).

Within the *Refreshing* paradigm, Ensemble batch prediction intervals (EnbPI) (Xu and Xie, 2023a) pioneered the use of bootstrap resampling and ensemble learning to sequentially construct prediction intervals without requiring data splitting. Sequential Predictive Conformal Inference (SPCI) (Xu and Xie, 2023b) further extends the EnbPI framework by leveraging Quantile Random Forests (QRF) to directly estimate conditional quantiles. Crucially, to handle asymmetric and skewed residual distributions in 1D sequences, SPCI introduces a dynamic tail-snipping mechanism; instead of assuming symmetric bounds, it optimizes an asymmetry parameter $\beta \in [0, \alpha]$, computing the final interval using the β -th and $(1 - \alpha + \beta)$ -th empirical quantiles of the non-conformity scores.

Transitioning these sequential adaptation techniques to multivariate spatial outputs introduces significant geometric complexity. The most naive approach constructs multidimensional regions by applying 1D CP independently across each spatial axis. To theoretically guarantee a joint valid coverage of $1 - \alpha$ across d dimensions, this independent approach relies on the Bonferroni correction, evaluating each marginal non-conformity score at a strictly tightened significance level of α/d . For 2D spatial coordinates, this effectively forces the 1D algorithms to target $1 - (\alpha/2)$. While this guarantees coverage, ignoring the spatial covariance intrinsically results in excessively conservative (inflated) rectangular bounding boxes that are suboptimal for tight physical environments.

To natively capture joint multidimensional distributions, advanced methods construct multivariate non-conformity measures using techniques such as Copula functions (Messoudi et al., 2021) to model the joint cumulative distribution, Mahalanobis distance (Johnstone and Cox, 2021; Xu et al., 2024) for covariance-aware geometric scaling. More recently, MultiDimSPCI (Xu et al., 2024) has extended the SPCI framework to multivariate time series by jointly calibrating ensemble-based multidimensional residuals. While mathematically sound and theoretically robust, these joint frameworks typically necessitate intensive offline calibration phases, complex density estimations, matrix inversions, or heavy recursive ensemble updates. Consequently, they inherently introduce substantial computational overhead. There is, therefore, a pressing need to refine multidimensional CP frameworks to be significantly more lightweight and computationally efficient. Bridging this gap to enable practical, real-time online deployment in time-sensitive applications forms the primary motivation for this work.

Uncertainty Quantification in Trajectory Forecasting: In the ADAS and autonomous driving domains, capturing the predictive uncertainty of dynamic agents is crucial. Traditional methods often employ Bayesian Neural Networks (BNNs) (Liu et al., 2022), Gaussian Processes (Mercat et al., 2019), or generative models like Conditional Variational Autoencoders (CVAEs) (Lee et al., 2017) to output probability distributions over future paths. However, these techniques typically require specific distributional assumptions and do not provide exact distribution-free coverage guarantees. When CP is applied to trajectory prediction to secure distribution-free bounds, it is frequently implemented by applying 1D CP independently to the x and y coordinates. As demonstrated in our experiments, this independent assumption ignores spatial covariance, leading to structurally invalid or excessively conservative regions during complex maneuvers.

Our Lightweight Online Multivariate CP framework bridges these disparate domains. By introducing an angular phase-space EMA for aspect ratio adaptation, it uniquely provides the online agility of sequential CP and the spatial awareness of multivariate methods, all while strictly adhering to the low-latency computational requirements of ADAS trajectory planners.

2.2. Standard Conformal Prediction

Conformal Prediction is a statistical framework that provides finite-sample, distribution-free validity guarantees. Consider a sequence of random variables $Z_1, \dots, Z_N$ where $Z_i = (X_i, Y_i) \in \mathcal{X} \times \mathcal{Y}$. Given a new test covariate X_{N+1} , the objective is to construct a confidence region $\hat{C}(X_{N+1}) \subseteq \mathcal{Y}$ that satisfies the marginal coverage property, as defined in

Equation (1):

$$\mathbb{P}(Y_{N+1} \in \hat{C}(X_{N+1})) \geq 1 - \alpha, \quad (1)$$

where $\alpha \in (0, 1)$ is a user-specified miscoverage rate. This is achieved by defining a non-conformity measure, a scalar score function $s_i = \mathcal{A}(Z_i, \mathcal{D}_{\text{cal}}) \in \mathbb{R}$, which quantifies the discrepancy between a point and the underlying data distribution through the calibration set $\mathcal{D}_{\text{cal}}$. The prediction region $\hat{C}(X_{N+1})$ is then formally constructed by including all candidate labels $y \in \mathcal{Y}$ for which the corresponding non-conformity score, denoted as $s(X_{N+1}, y)$, falls below or equals the $(1 - \alpha)$ -th empirical quantile of the calibration scores. Mathematically, this set is defined in Equation (2):

$$\hat{C}(X_{N+1}) = \{y \in \mathcal{Y} \mid s(X_{N+1}, y) \leq \hat{q}_{1-\alpha}\}, \quad (2)$$

where $\hat{q}_{1-\alpha}$ represents the $(1 - \alpha)$ -th empirical quantile computed from the calibration set $\mathcal{D}_{\text{cal}}$. Crucially, this rigorous theoretical guarantee is strictly dependent upon the assumption of *exchangeability*. A sequence of random variables is infinitely exchangeable if their joint probability distribution is invariant under any permutation π of their indices, expressed in Equation (3):

$$P(Z_1, \dots, Z_N) = P(Z_{\pi(1)}, \dots, Z_{\pi(N)}). \quad (3)$$

2.3. Conformal Prediction for Time Series Data

In real-time ADAS applications, trajectory observations are presented as sequential time-series data $(X_t, Y_t)_{t=1}^{\infty}$. The inherent temporal auto-correlations and dynamic distributional shifts explicitly violate the exchangeability assumption, causing standard CP to lose its theoretical validity.

To handle these non-stationary environments, Conformal Forecasting algorithms introduce continuous calibration mechanisms. Specifically, to address sudden distribution shifts, Adaptive Conformal Inference (ACI) (Gibbs and Candes, 2021) treats the problem as an online control task. Rather than relying on static error rates, ACI continuously monitors recent predictive performance and dynamically updates the nominal significance level α_t at each time step t . This is achieved using the recursive equation shown in Equation (4) to target the desired $1 - \alpha$ coverage:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad (4)$$

where $\gamma > 0$ is the learning rate (step size), and $\text{err}_t = \mathbf{1}\{Y_t \notin \hat{C}_{t-1}\}$ is the instantaneous miscoverage indicator. The method dynamically adjusts α_t based on immediate error feedback, increasing the error margin when coverage is met and tightening it upon miscoverage. This mechanism ensures that the empirical coverage robustly converges to the target $1 - \alpha$ level over time, effectively adapting to the instability of dynamic trajectories.

It is important to note that because the non-conformity score inherently reduces any multi-dimensional residual to a scalar $s_i \in \mathbb{R}$, adaptive algorithms like ACI are theoretically agnostic to the dimensionality of $\mathcal{Y}$. Consequently, the primary bottleneck in extending CP to multivariate spatial data lies not only in the sequential adaptation mechanisms, but rather in formulating a geometrically sound, lightweight joint non-conformity measure.

2.4. Problem Setup and Probabilistic Guarantees

We consider a multi-dimensional sequential forecasting setup. For a discrete time index $t = 1, 2, \dots$, observations $Z_t = (X_t, Y_t)$ are sequentially revealed. In the specific context of ADAS trajectory prediction, $Y_t \in \mathbb{R}^2$ represents the impending 2D spatial coordinates of the target agent. The feature vector $X_t \in \mathcal{X}$ encapsulates all information available prior to time t . Specifically, to capture the inherent temporal dynamics, X_t is formulated by explicitly incorporating a finite historical window of the past trajectory states, defined as $X_t = (Y_{t-H}, Y_{t-H+1}, \dots, Y_{t-1})$, where H denotes the look-back window size. Crucially, we allow for an arbitrarily unknown correlation structure among the sequential observations Z_t , explicitly reflecting the non-stationary and dynamically shifting nature of physical driving environments.

The primary objective is to sequentially construct a confidence region $\hat{C}_{t-1}(X_t) \subseteq \mathbb{R}^2$ for the upcoming time step t . This region depends on historical observations and a user-specified significance level $\alpha \in (0, 1)$. The fundamental requirement is that this region must encapsulate the unobserved ground truth Y_t with a probability of at least $1 - \alpha$ within minimum confidence region size.

When evaluating the reliability of these regions sequentially over time, the strength of the mathematical guarantee must be precisely defined. There are two primary types of coverage to be satisfied by $\hat{C}_{t-1}(X_t)$:

- **Marginal Coverage:** The weaker form of validity, which ensures that the error rate is bounded by α when averaged over the entire joint distribution of the sequence. As formulated in Equation (5):

$$\mathbb{P}(Y_t \in \hat{C}_{t-1}(X_t)) \geq 1 - \alpha, \quad \forall t \quad (5)$$

If $\hat{C}_{t-1}(X_t)$ satisfies Equation (5), it is considered marginally valid. While mathematically sound globally, marginal coverage may mask localized periods of systemic failure, such as severe under-coverage during complex maneuvering that is statistically masked by over-coverage during simple, straight-line driving.

- **Conditional Coverage:** A significantly stronger theoretical condition requiring the coverage guarantee to hold strictly conditioned on the specific input state X_t . As defined in Equation (6):

$$\mathbb{P}(Y_t \in \hat{C}_{t-1}(X_t) \mid X_t) \geq 1 - \alpha, \quad \forall t \quad (6)$$

If the region satisfies Equation (6), it is conditionally valid. In sequential trajectory forecasting, achieving exact conditional coverage in continuous spaces is mathematically impossible without strict distributional assumptions. Therefore, state-of-the-art frameworks aim to achieve strong *empirical local coverage* over time, ensuring the $(1 - \alpha)$ guarantee holds robustly within tightly localized temporal windows regardless of the instantaneous driving context.

Once $\hat{C}_{t-1}(X_t)$ rigorously satisfies these target validity guarantees, the secondary objective is to construct regions that are as small as geometrically possible. Minimizing this spatial footprint is essential to quantify uncertainty precisely and provide tight, actionable bounds for autonomous planning modules.

2.5. Evaluation Metrics

To rigorously quantify the performance of the proposed online multivariate framework, we evaluate the generated confidence regions across two fundamental and inherently competing metrics:

- **Validity (Empirical Coverage):** Reflects the statistical reliability of the framework. We evaluate this at both the marginal and conditional levels. The global marginal validity over a sequence of length T is defined in Equation (7):

$$V_{\text{global}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{Y_t \in \hat{C}_{t-1}\} \quad (7)$$

To assess local conditional coverage, validity is continuously computed over a sliding temporal window of size W , as shown in Equation (8):

$$V_{\text{local}}(t) = \frac{1}{W} \sum_{\tau=t-W+1}^t \mathbf{1}\{Y_{\tau} \in \hat{C}_{\tau-1}\} \quad (8)$$

A theoretically sound framework must maintain $V_{\text{global}} \geq 1 - \alpha$ while minimizing the variance of $V_{\text{local}}(t)$ across the entire temporal sequence.

- **Efficiency (Region Size):** A trivial algorithm could achieve perfect validity by generating infinitely large regions, rendering them useless for ADAS planning modules. Therefore, efficiency is quantified by the spatial area of the generated 2D confidence region $\hat{C}_t$, detailed in Equation (9):

$$\text{Efficiency} = \frac{1}{T} \sum_{t=1}^T \text{Area}(\hat{C}_{t-1}) \quad (9)$$

An optimal multivariate CP framework minimizes this spatial area to provide tight, highly actionable bounds without violating the validity constraints.

- **Computational Latency (Inference Time):** For real-time ADAS applications, ultra-low latency is a strict operational constraint. A framework is only viable if it can dynamically update and generate confidence regions significantly faster than the environmental sampling rate. Therefore, we evaluate computational efficiency empirically by measuring the average execution latency per online update iteration. A truly lightweight framework must demonstrate minimal computational overhead, ensuring the prediction cycle operates well within the system’s real-time execution limits.

3. Methodology

The primary objective of the proposed framework is to extend the standard Conformal Prediction (CP) methodology from univariate point forecasting to a multivariate, two-dimensional (2D) space. Specifically designed for Advanced Driver Assistance Systems

(ADAS), this approach constructs 2D prediction regions for the future trajectories of dynamic objects. By directly tackling the spatial variance for each target while maintaining an ultra-low computational footprint, the algorithm provides autonomous planning modules with rigorous uncertainty bounds crucial for safe and proactive decision-making.

3.1. Joint 2D Non-Conformity Measure Formulation

Instead of treating the longitudinal (x) and lateral (y) axes independently, our method calibrates the multidimensional data jointly. To achieve this, we extract a unified feature that intrinsically links the predictive errors across all dimensions. Within the 2D spatial domain, we utilize the empirical aspect ratio as the primary joint feature.

For a predicted trajectory point $(\hat{x}_i, \hat{y}_i)$ and its corresponding ground truth (x_i, y_i) , let the absolute prediction errors be denoted as $e_{x,i} = |\hat{x}_i - x_i|$ and $e_{y,i} = |\hat{y}_i - y_i|$. By evaluating the mean of these aspect ratios, denoted as a_{mean} , across a calibration set, we isolate the dominant directional trend—or majority direction—of the prediction errors.

Building upon a_{mean} , we formally define the 2D non-conformity score function. we construct a base bounding box that exhibits this empirical aspect ratio while maintaining a normalized unit area, its dimensions are naturally factored by the square root of the aspect ratio ($\sqrt{a_{\text{mean}}}$). As visualized in Figure 1, the objective is to determine a universal scaling factor that dilates this base unit-area rectangle that centered at the origin with half-width $1/\sqrt{a_{\text{mean}}}$ and half-height $\sqrt{a_{\text{mean}}}$ to completely enclose the spatial error vector.

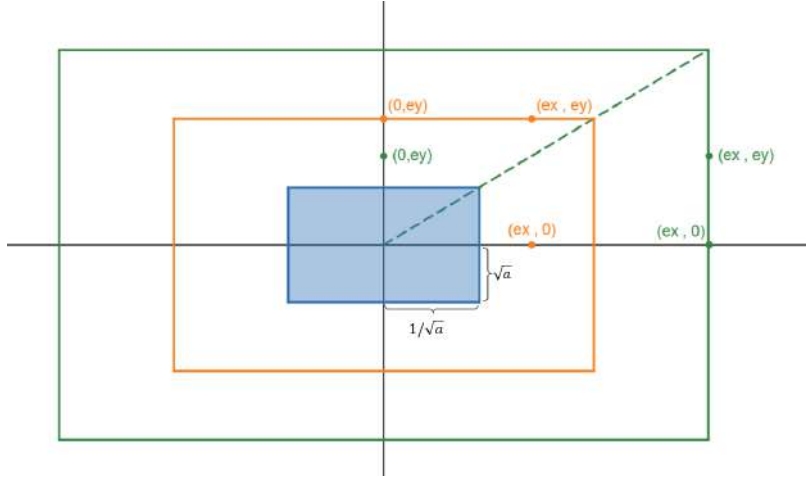


Figure 1: Geometric formulation of the 2D non-conformity measure. The score s_i represents the minimal scaling factor t required to dilate a base unit-area rectangle (proportioned by $\sqrt{a_{\text{mean}}}$) until it fully encloses the spatial error $(e_{x,i}, e_{y,i})$.

Mathematically, the non-conformity score s_i for the i -th observation is defined as the infimum of all possible scaling factors t that can simultaneously bound these dimensionally

normalized errors. This is formally expressed in Equation (10):

$$s_i = \inf \left\{ t : t \geq \sqrt{a_{\text{mean}}} \cdot e_{x,i}, t \geq \frac{e_{y,i}}{\sqrt{a_{\text{mean}}}} \right\}. \quad (10)$$

By evaluating this infimum, the framework geometrically identifies whether the longitudinal or lateral error acts as the active constraint (as shown by the orange and green scaling boundaries in Figure 1). This infimum formulation naturally simplifies to evaluating the maximum of the two scaled error terms, as shown in Equation (11):

$$s_i = \max \left(\sqrt{a_{\text{mean}}} \cdot e_{x,i}, \frac{e_{y,i}}{\sqrt{a_{\text{mean}}}} \right). \quad (11)$$

By selecting this maximum norm, the framework guarantees that scaling the bounding box to accommodate the maximum active constraint intrinsically ensures that the complementary dimension is also strictly covered.

3.2. Robust Online Adaptation Mechanism

In a dynamic ADAS environment, the statistical distribution of trajectory prediction errors is rarely stationary. To maintain the robustness of the confidence regions in the presence of distribution shifts, the error direction a_{mean} must dynamically adapt to the most recent trajectory observations.

A standard approach we propose for online parameter updating is the Exponential Moving Average (EMA), which could be formulated as in Equation (12):

$$a_{\text{mean}}^{(t)} = \lambda a_{\text{mean}}^{(t-1)} + (1 - \lambda) \left(\frac{e_{y,t}}{e_{x,t}} \right). \quad (12)$$

However, this linear update mechanism is highly vulnerable to outliers. Specifically, if the longitudinal error $e_{x,t}$ infinitesimally approaches zero, the instantaneous aspect ratio trends toward infinity. Such extreme values would disproportionately skew the EMA, leading to artificially inflated, unstable prediction regions.

To systematically resolve this numerical instability, we propose projecting the aspect ratio into a bounded angular phase space prior to aggregation. By applying the inverse tangent function ($\arctan$), the unbounded domain of the error ratio is mathematically constrained to a strictly bounded angular range of $[0, \pi/2]$ for absolute errors. This non-linear transformation acts as a natural stabilizer, suppressing the magnitude of extreme outliers while preserving the true geometric relationship. For all subsequent online updates, governed by the adaptiveness factor $\lambda \in (0, 1)$, the robust online update mechanism is formally defined in Equation (13):

$$a_{\text{mean}}^{(t)} = \tan \left[\lambda \arctan \left(a_{\text{mean}}^{(t-1)} \right) + (1 - \lambda) \arctan \left(\frac{e_{y,t}}{e_{x,t}} \right) \right], \quad (13)$$

During the initialization phase over the first historical window of size H , the baseline aspect ratio is computed by averaging these projected angles and mapping the result back to the ratio space using the tangent function, as detailed in Equation (14):

$$a_{\text{mean}} = \tan \left(\frac{1}{H} \sum_{i=1}^H \arctan \left(\frac{e_{y,i}}{e_{x,i}} \right) \right). \quad (14)$$

3.3. Pseudo-code and Implementation

Having established the theoretical foundations, this section synthesizes the joint non-conformity measure and the robust angular phase-space adaptation into a unified operational framework. The architecture is further enhanced with Adaptive Conformal Inference (ACI) to provide dynamic threshold calibration. The proposed framework operates sequentially, taking as functional inputs a continuous data stream of 2D trajectory coordinates (x_t, y_t) and a base machine learning or kinematic model $\hat{f}$ for point predictions. Its execution is strictly governed by four user-defined hyperparameters: the target significance level $\alpha \in (0, 1)$, the EMA adaptiveness factor λ , the ACI learning rate γ , and the historical window size H .

At each time step t , the algorithm outputs a calibrated 2D geometric confidence region $\hat{C}_{t-1}(X_t)$. This region is formulated as a dynamically scaled rectangle mathematically guaranteed to encapsulate the true future trajectory point. The complete execution logic is formalized in Algorithm 1.

The operational flow of Algorithm 1 begins with an initial baseline calibration sequence. During the preliminary window of size H , the algorithm generates historical predictions utilizing the base model and calculates the absolute spatial errors against the ground truth. By projecting these initial error aspect ratios into the angular phase space, calculating the mean, and mapping it back, the baseline error direction a_{mean} is strictly established. This formulation ensures numerical stability against initial outliers before the continuous streaming sequence commences.

Once the baseline parameters are established, the system transitions to a continuous online processing loop. For each newly received trajectory observation, the execution follows four critical steps:

1. **Adaptation:** The most recent spatial errors are appended to the active memory sets. The angular EMA dynamically updates a_{mean} , smoothly integrating the latest geometric error ratio. The adaptiveness factor λ guarantees that transient mathematical spikes do not persistently destabilize the region’s geometry.
2. **Evaluation:** The algorithm evaluates the joint 2D non-conformity measure across all H points currently within the active sliding window. Utilizing the square-root scaled maximum norm, this computationally lightweight step rapidly identifies the necessary s_j scores.
3. **Region Construction:** The optimal scaling threshold s^* is extracted by computing the $(1 - \alpha_t)$ empirical quantile of the evaluated scores. The final prediction region is explicitly generated by projecting s^* through the square root of the aspect ratio. This yields strictly calibrated half-width $(s^*/\sqrt{a_{\text{mean}}})$ and half-height $(s^* \cdot \sqrt{a_{\text{mean}}})$ constraints, forming a robust bounding box centered on the heuristic prediction.
4. **Feedback and Constraint Enforcement:** Finally, the ACI feedback mechanism evaluates whether the true coordinate successfully fell within the newly generated boundaries. It updates the target significance α_{t+1} based on the predefined recursive rule in Equation (4). The oldest errors are subsequently dequeued from the calibration sets to enforce the temporal sliding window constraint, guaranteeing that the statistical properties remain tightly coupled to the latest driving dynamics.

Algorithm 1: Lightweight Online Multivariate Conformal Prediction (L-OMCP)

Input: Data stream (x_t, y_t) , base model $\hat{f}$, initial significance α , adaptiveness λ , ACI rate γ , window size H .

Output: Confidence region $\hat{C}_{t-1}(X_t)$ for each point prediction

Initialize $\hat{e}_x, \hat{e}_y \leftarrow \emptyset$ as empty queues of max size H ;

Set $\alpha_1 \leftarrow \alpha$;

for $h \leftarrow 1$ **to** H **do**

Compute $\hat{x}_h, \hat{y}_h \leftarrow \hat{f}(x_{-H}, y_{-H})$;

$\hat{e}_x \leftarrow \hat{e}_x \cup \{|x_h - \hat{x}_h|\}$, $\hat{e}_y \leftarrow \hat{e}_y \cup \{|y_h - \hat{y}_h|\}$;

end

$a_{\text{mean}} \leftarrow \tan \left(\frac{1}{H} \sum_{i=1}^H \arctan \left(\frac{\hat{e}_y^i}{\hat{e}_x^i} \right) \right)$ Equation (14) ;

foreach point (x_t, y_t) **do**

Compute $\hat{x}_t, \hat{y}_t \leftarrow \hat{f}(x_{<t}, y_{<t})$;

for $j \leftarrow 1$ **to** H **do**

$s_j \leftarrow \max \left(\sqrt{a_{\text{mean}}} \cdot \hat{e}_x^j, \frac{\hat{e}_y^j}{\sqrt{a_{\text{mean}}}} \right)$;

end

$s^* \leftarrow (1 - \alpha_t)$ Quantile of s ;

$\hat{C}_{t-1}(x_t, y_t) \leftarrow \text{rectangle} \left((\hat{x}_t, \hat{y}_t), \frac{s^*}{\sqrt{a_{\text{mean}}}}, s^* \cdot \sqrt{a_{\text{mean}}} \right)$;

$\alpha_{t+1} \leftarrow \alpha_t + \gamma(\alpha - \mathbf{1}\{y_t \notin \hat{C}_{t-1}(X_t)\})$;

$e_{x,t} \leftarrow |x_t - \hat{x}_t|$, $e_{y,t} \leftarrow |y_t - \hat{y}_t|$;

$\hat{e}_x \leftarrow \hat{e}_x \cup \{e_{x,t}\}$, $\hat{e}_y \leftarrow \hat{e}_y \cup \{e_{y,t}\}$;

$a_{\text{mean}} \leftarrow \tan \left[\lambda \arctan(a_{\text{mean}}) + (1 - \lambda) \arctan \left(\frac{e_{y,t}}{e_{x,t}} \right) \right]$ Equation (13) ;

Remove oldest element from $\hat{e}_x$, $\hat{e}_y$;

return $\hat{C}_{t-1}(X_t)$;

end

4. Experimental Setup

To evaluate the L-OMCP framework, we conducted comprehensive experiments across synthetic and real-world scenarios. Our experimental design highlights the empirical validity of the joint 2D calibration and validates the adaptive mechanisms. Furthermore, it benchmarks computational and statistical performance against a Bonferroni-corrected independent 1D baseline and the MultiDimSPCI multivariate method.

4.1. Datasets

We empirically validate our framework across two distinct domains: statistical benchmarking and real-world automotive application.

4.1.1. STATISTICAL BENCHMARKS: SYNTHETIC AND WIND DATA

Synthetic Trajectory: A 7-minute driving sequence (10 Hz, 4,200 states) featuring aggressive lane changes and 25 m/s sinusoidal longitudinal velocity. To simulate prediction degradation, we injected AR(1) noise and maneuver-induced lag proportional to the lateral gradient into the base heuristic (Figure 2). This isolates the framework’s ability to recover coverage during sudden distribution shifts.

Bivariate Wind Data: Aligning with multivariate CP literature, we utilize the standard 2D wind speed dataset to benchmark against algorithms like MultiDimSPCI. Exhibiting severe heteroscedasticity and cross-dimensional correlation, this time-series provides a rigorous sandbox to validate our formulation independent of automotive applications.

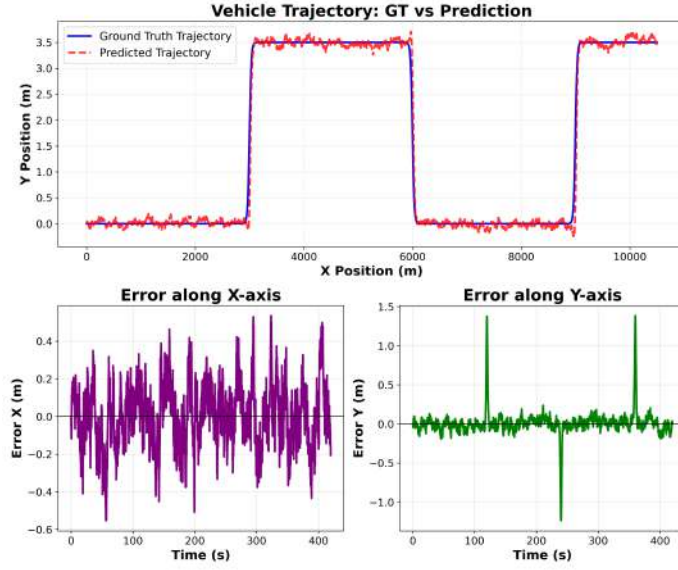


Figure 2: Overview of the synthetic driving sequence. The plots display the continuous lateral trajectory ($Y(m)$ vs. $X(m)$) exhibiting multiple lane changes, alongside the corresponding independent spatial prediction residuals (e_x and e_y).

4.1.2. ADAS VALIDATION: REAL-WORLD RENAULT DATASET

To validate against complex physical constraints, we utilize a 5-minute proprietary Renault trajectory (33.3 Hz, 10,000 states). This diverse scenario includes sharp accelerations, successive lane changes, and an urban-to-highway transition.

Reflecting a purely data-driven pipeline, base point predictions ($\hat{f}$) were generated using a learning-based model trained on historical ego-centric sequences. L-OMCP operates as the supervisory top layer, actively quantifying uncertainty to dynamically construct risk-aware confidence regions around these predictions (Figure 3).

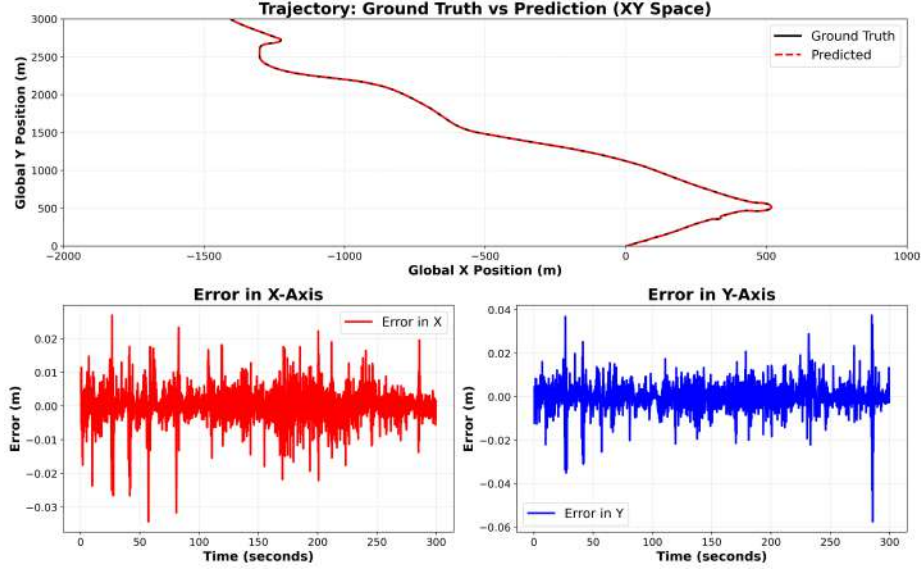


Figure 3: Overview of the Real-World Renault dataset. The plots depict the 2D spatial trajectory ($Y(m)$ vs. $X(m)$) of the dynamic agent across complex urban and highway environments, alongside the tracking residuals (e_x and e_y).

4.2. Benchmarks and Baselines

To demonstrate the joint multivariate online approach in terms of both statistical performance and computational latency, the proposed L-OMCP algorithm is benchmarked against two primary paradigms:

- Independent 1D CP (Baseline):** In this approach, standard univariate sequential Conformal Prediction is applied independently to the longitudinal (x) and lateral (y) dimensions. To guarantee a joint valid coverage of $1 - \alpha$, this baseline enforces the Bonferroni correction, targeting a strictly tightened marginal significance level of $\alpha/2$ for each dimension before taking the Cartesian product of the resulting 1D intervals. Furthermore, we integrate this method with Adaptive Conformal Inference (ACI) to establish a fair benchmark against our proposed L-OMCP method.
- MultiDimSPCI:** We benchmark against Multidimensional Sequential Predictive Conformal Inference, a framework designed for multivariate time-series data (Xu et al., 2024). This comparison strictly evaluates our lightweight formulation against computationally heavier multivariate conformal structures, demonstrating that L-OMCP achieves reliable performance while operating with a significantly lower computational overhead.

4.3. Results

In this section, we present the empirical performance of our proposed L-OMCP framework alongside the benchmarked methods. For all experiments, the target miscoverage rate is set to $\alpha = 0.10$, corresponding to a nominal target coverage of 90%. Furthermore, the hyperparameters are configured with $\gamma = 0.10$, and $\lambda = 0.9$.

We first evaluate the overall statistical performance and computational overhead across the entire sequential horizons. Table 1 summarizes the marginal empirical coverage, the average spatial area of the generated confidence regions, and the average inference latency per iteration for the considered datasets,

Table 1: Performance comparison of the proposed framework on Synthetic, Renault, and Wind datasets at $\alpha = 0.10$.

Method	Synthetic Data			Renault Data			Wind Data		
	Cov (%)	Area (m ²)	Time (ms)	Cov (%)	Area (m ²)	Time (ms)	Cov (%)	Area	Time (ms)
1D_Independent	88.7	0.2773	0.323	89.2	0.4342	0.407	90.1	2.1343	0.319
MultiDimSPCI	89.8	0.2607	264.0	85.9	0.3426	381.6	88.5	1.8249	159.3
L-OMCP (Ours)	88.6	0.1679	0.199	88.9	0.2921	0.229	89.6	1.7507	0.175

As observed in Table 1, **1D_Independent** exhibits the expected behavior of a Bonferroni-calibrated baseline. If both marginal intervals are valid at level $1 - \alpha/2$, the union bound gives $\mathbb{P}(Y_t \notin C_{x,t} \times C_{y,t}) \leq \alpha$, irrespective of the dependence between coordinates. Consistent with this construction, its empirical joint coverage remains close to the nominal 90% level, reaching 88.7%, 89.2%, and 90.1% on Synthetic, Renault, and Wind, respectively. This coverage-oriented behavior comes at the cost of conservative regions: the baseline produces the largest average area on every dataset (0.2773 m², 0.4342 m², and 2.1343, respectively), illustrating the unused space introduced by independently expanding both coordinate-wise intervals. **MultiDimSPCI** reduces these areas, but its coverage is less consistent across the datasets (89.8%, 85.9%, and 88.5%) and its online Quantile Regression Forest queries incur 159.3–381.6 ms of latency. In contrast, **L-OMCP** retains coverage close to both the nominal level and the Bonferroni reference while producing the smallest average regions in all three experiments. Relative to **1D_Independent**, its coverage differs by only 0.1, 0.3, and 0.5 percentage points, whereas its average area decreases by 39.5%, 32.7%, and 18.0% on Synthetic, Renault, and Wind, respectively. L-OMCP also requires only 0.175–0.229 ms per update, reducing latency by 38–45% relative to the independent baseline and yielding an approximately 900–1,700-fold speedup over **MultiDimSPCI**. These results demonstrate a favorable empirical coverage–area–latency trade-off: L-OMCP preserves nearly the same coverage as the conservative Bonferroni construction while providing substantially tighter regions with negligible computational overhead.

4.4. Temporal Dynamics and Local Adaptability

The temporal results support this aggregate comparison. On the Synthetic dataset (Figure 4), the sliding-window coverage of **1D_Independent** generally evolves around the 90% target, as expected from the Bonferroni construction. L-OMCP follows a similarly target-

centered profile while maintaining a markedly smaller region-area distribution. Although all methods enlarge their regions during abrupt maneuvers, the most pronounced event produces a larger and more persistent expansion for **MultiDimSPCI**.

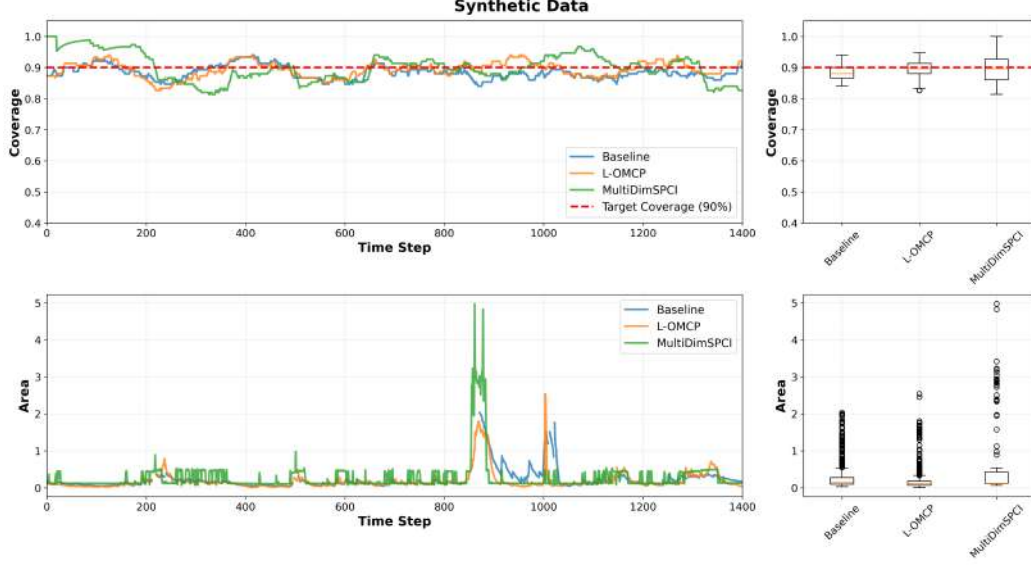


Figure 4: Temporal evaluation on the Synthetic Dataset.

By adapting the rectangle’s aspect ratio to recent directional errors, L-OMCP accommodates these changes without uniformly inflating both coordinate-wise bounds, yielding a 39.5% average-area reduction with only a 0.1-percentage-point coverage difference from the baseline. A similar pattern appears on the real-world Renault trajectory (Figure 5): L-OMCP remains near the nominal coverage level and closely follows the Bonferroni baseline, whereas **MultiDimSPCI** exhibits larger departures below the target. The ACI feedback adjusts the threshold s^* in response to recent errors, while the angular EMA reallocates it between the spatial directions, limiting unnecessary geometric inflation. This produces Renault area reductions of 32.7% relative to **1D_Independent** and 14.7% relative to **MultiDimSPCI**. The Wind results show the same trade-off: L-OMCP attains 89.6% coverage versus 90.1% for the baseline while reducing average area by 18.0%. Across all three sequences, L-OMCP therefore improves spatial efficiency and responsiveness without materially sacrificing empirical coverage.

4.5. Ablation Study

A core contribution of our framework is its dynamic adaptiveness. To empirically isolate and quantify the impact of our chosen hyperparameters on the geometric stability and validity of the bounding regions, we conducted two targeted ablation studies:

- **Impact of the ACI Learning Rate (γ):** By varying γ , which controls the coverage feedback aggressiveness, we identified the optimal fixed rate to balance statistical

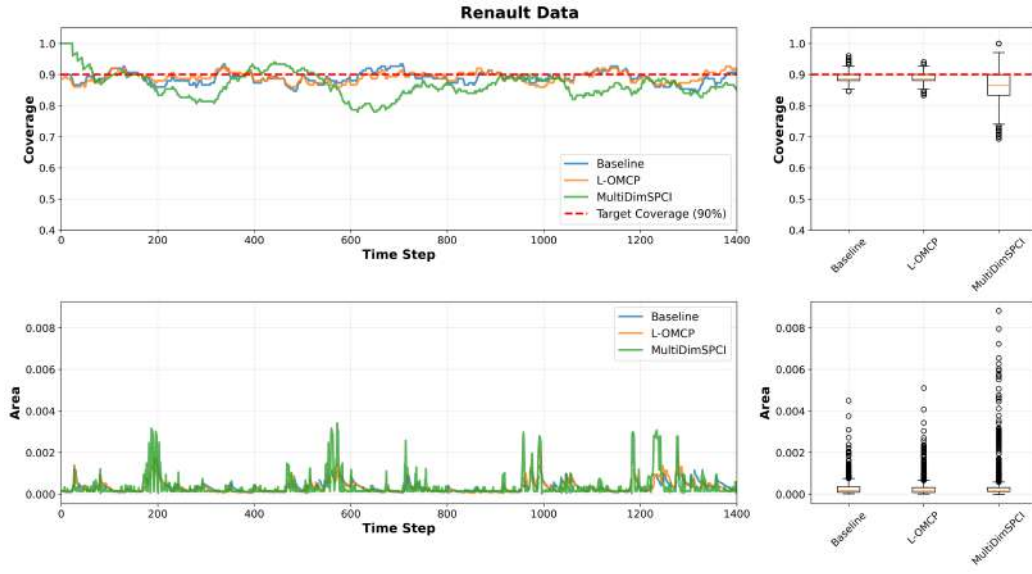


Figure 5: Temporal evaluation on the real-world Renault Dataset.

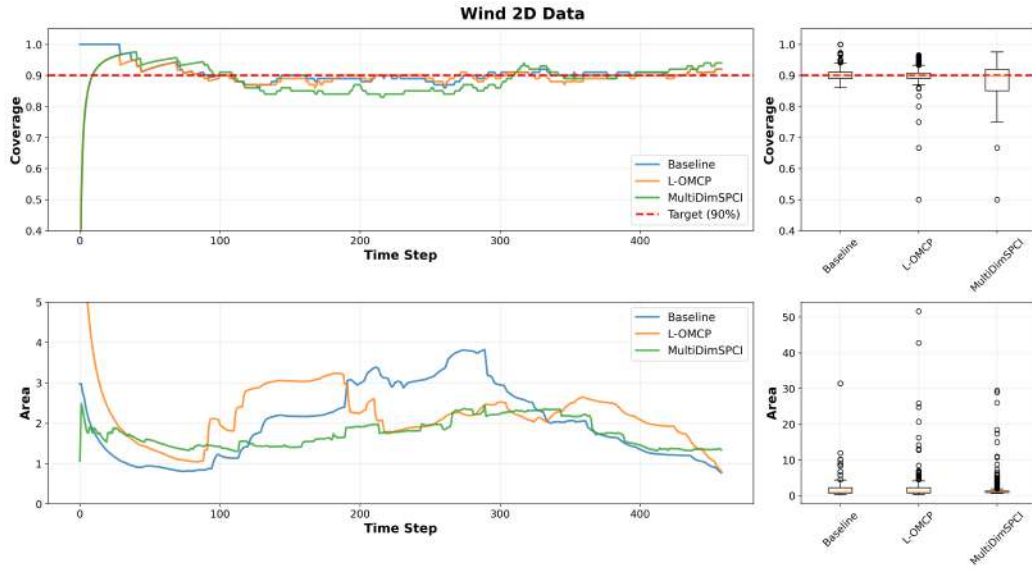


Figure 6: Temporal evaluation on the real-world Wind Dataset.

stability with rapid recovery from distribution shifts (Figure 7). While advanced meta-algorithms like AgACI (Zaffran et al., 2022) dynamically optimize γ , they introduce extra computational complexity. To preserve our framework’s lightweight design, we utilize a heuristically optimized fixed γ , reserving dynamic expert aggregation for future work.

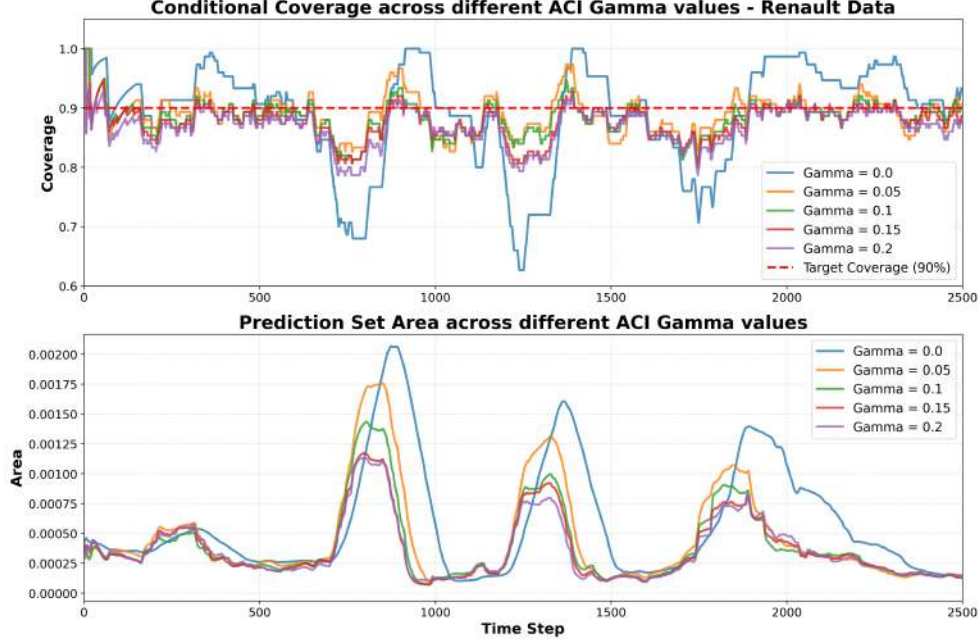


Figure 7: Sensitivity analysis of the ACI learning rate (γ).

- **Impact of the EMA Adaptiveness Factor (λ):** The λ parameter governs the EMA update, dictating the geometric responsiveness of our non-conformity measure. As illustrated in Figure 8, by varying λ , we isolate the core trade-off between structural stability against sensor noise (higher λ) and agility during sudden maneuvers like lane changes (lower λ). This validates our selected configuration, ensuring robust kinematic tracking without transient jitter.

5. Conclusion

This paper presented L-OMCP, a lightweight framework for online multivariate conformal prediction under the latency constraints of ADAS trajectory forecasting. It uses a joint 2D non-conformity measure whose geometry is determined by an online estimate of the empirical error aspect ratio. The angular phase-space exponential moving average stabilizes the region geometry against extreme error ratios, while Adaptive Conformal Inference updates the conformal threshold under distribution shifts. L-OMCP therefore adapts the allocation of uncertainty between the two coordinates according to their recent directional error

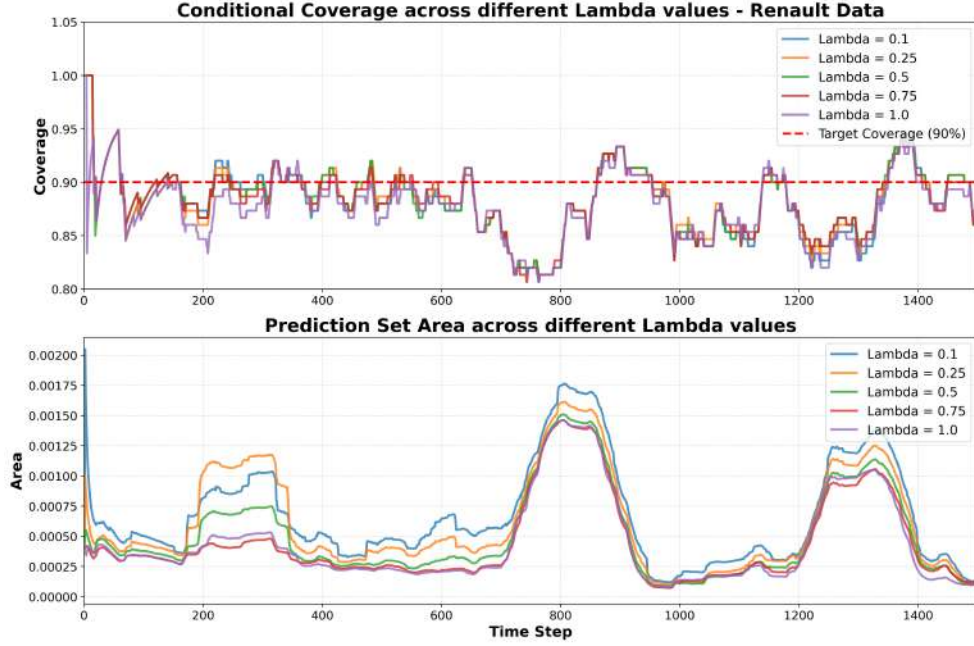


Figure 8: Sensitivity analysis of the Adaptiveness factor (λ).

imbalance, without requiring the matrix operations or auxiliary learning-based estimators associated with more computationally demanding multivariate approaches.

Experiments on the Synthetic, Renault, and Wind datasets demonstrate a favorable empirical coverage–area–latency trade-off. L-OMCP achieves 88.6–89.6% joint coverage, remaining close to the nominal 90% level and to the Bonferroni- corrected independent baseline, while reducing the baseline’s average region area by 18.0–39.5%. Compared with **MultiDimSPCI**, L-OMCP produces smaller average regions across all three datasets and reduces the inference latency by approximately 900–1,700 times, while exhibiting more consistent coverage on the Renault and Wind datasets. These results show that L-OMCP provides tight and computationally efficient prediction regions without materially sacrificing empirical coverage. Future research will focus on extending this geometric formulation from single-point predictions to multi-step forecasting horizons. By constructing cohesive confidence tubes across the entire predicted trajectory, we aim to provide joint coverage guarantees that are essential for safe, proactive, and risk-aware downstream motion planning.

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